EMM 307

CAT 1

1 hr

Answer all questions

Q1. A steam power plant operates between a boiler pressure of 42 bar and a condenser of 0.035 bar. Calculate for these limits:

(i) the cycle efficiency
(ii) the work ratio and
(iii) the specific steam consumption

a) for a carnot cycle using wet steam
b) for a rankine cycle with dry saturated steam at entry to the turbine: and
c) the rankine cycle of (b) when the expansion process has an isentropic efficiency of 80%
d) when in (b) steam is superheated to 500^0C neglect feed pump work
e) Calculate the new cycle efficiency and specific steam consumption if reheat is included in the plant.

The steam conditions at inlet to the turbine are 42 bar, 500^0C and the condenser pressure is 0.035 bar as before.

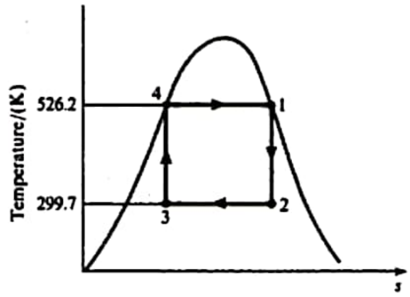
Assume that the steam just dry saturated on leaving the first turbine and is reheated to its initial temperature. Neglect the feed pump term. (20 marks)

Example 8.1 A steam power plant operates between a boiler pressure of 42 bar and a condenser pressure of 0.035 bar. Calculate for these limits the cycle efficiency, the work ratio, and the specific steam consumption:

(i) for a Carnot cycle using wet steam;
(ii) for a Rankine cycle with dry saturated steam at entry to the turbine;
(iii) for the Rankine cycle of (ii), when the expansion process has an isentropic efficiency of 80%.

Solution (i) A Carnot cycle is shown in Fig. 8.5.

Fig. 8.5 Carnot cycle for Example 8.1(a)



$$T_1 = \text{saturation temperature at 42 bar}$$
$$= 253.2 + 273 = 526.2 \text{ K}$$
$$T_2 = \text{saturation temperature at 0.035 bar}$$
$$= 26.7 + 273 = 299.7 \text{ K}$$

Then from equation (5.1)

$$\eta_{\text{Carnot}} = \frac{T_1 - T_2}{T_1} = \frac{526.2 - 299.7}{526.2} = 0.432 \quad \text{or} \quad 43.2\%$$

Also Heat supplied $= h_1 - h_4 = h_{fg}$ at 42 bar $= 1698$ kJ/kg

Then $\eta_{\text{Carnot}} = \dfrac{\text{Net work output, } -\sum W}{\text{Gross heat supplied}} = 0.432$

Therefore $-\sum W = 0.432 \times 1698$,

i.e. Net work output, $-\sum W = 734$ kJ/kg

To find the gross work of the expansion process it is necessary to calculate h_2, using the fact that $s_1 = s_2$.
From tables

$$h_1 = 2800 \text{ kJ/kg} \quad \text{and} \quad s_1 = s_2 = 6.049 \text{ kJ/kg K}$$

Using equation (4.10)

$$s_2 = 6.049 = s_{f_2} + x_2 s_{fg_2} = 0.391 + x_2 8.13$$

therefore

$$x_2 = \frac{6.049 - 0.391}{8.13} = 0.696$$

Then using equation (2.2)

$$h_2 = h_{f_2} + x_2 h_{fg_2} = 112 + (0.696 \times 2438) = 1808 \text{ kJ/kg}$$

Hence, from equation (8.2)

$$-W_{12} = h_1 - h_2 = 2800 - 1808 = 992 \text{ kJ/kg}$$

Therefore we have, using equation (8.13),

$$\text{Work ratio} = \frac{\text{net work output}}{\text{gross work output}} = \frac{734}{992} = 0.739$$

Using equation (8.14)

$$\text{ssc} = \frac{1}{734}$$

i.e. $\text{ssc} = 0.001\,36$ kg/kW s

$= 4.91$ kg/kW h

(ii) The Rankine cycle is shown in Fig. 8.6.
As in part (i)

$$h_1 = 2800\ \text{kJ/kg} \quad \text{and} \quad h_2 = 1808\ \text{kJ/kg}$$

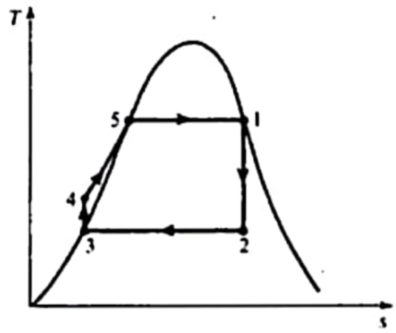
Also, $h_3 = h_f$ at 0.035 bar $= 112$ kJ/kg

Using equation (8.10), with $v = v_f$ at 0.035 bar

$$\text{Pump work} = v_f(p_4 - p_3) = 0.001 \times (42 - 0.035) \times \frac{10^5}{10^3}$$

$$= 4.2\ \text{kJ/kg}$$

Fig. 8.6 *T–s* diagram for Example 8.1(b)



Using equation (8.2)

$$-W_{12} = h_1 - h_2 = 2800 - 1808 = 992\ \text{kJ/kg}$$

Then using equation (8.8)

$$\eta_R = \frac{(h_1 - h_2) - (h_4 - h_3)}{(h_1 - h_3) - (h_4 - h_3)} = \frac{992 - 4.2}{(2800 - 112) - 4.2} = 0.368$$

i.e. $\eta_R = 36.8\%$

Using equation (8.13)

$$\text{Work ratio} = \frac{\text{net work output}}{\text{gross work output}} = \frac{992 - 4.2}{992} = 0.996$$

Using equation (8.14)

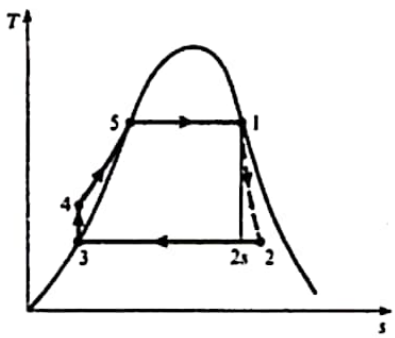
$$\text{ssc} = \frac{1}{-\sum W}$$

i.e. $\text{ssc} = \dfrac{1}{(992 - 4.2)} = 0.001\,01\ \text{kg/kW s} = 3.64\ \text{kg/kW h}$

(iii) The cycle with an irreversible expansion process is shown in Fig. 8.7.
Using equation (8.12)

$$\text{Isentropic efficiency} = \frac{h_1 - h_2}{h_1 - h_{2s}} = \frac{-W_{12}}{-W_{12s}}$$

Fig. 8.7 *T–s* diagram for Example 8.1(c)



therefore

$$0.8 = \frac{-W_{12}}{992}$$

i.e. $$-W_{12} = 0.8 \times 992 = 793.6 \text{ kJ/kg}$$

Then the cycle efficiency is given by

$$\text{Cycle efficiency} = \frac{(h_1 - h_2) - (h_4 - h_3)}{\text{gross heat supplied}}$$

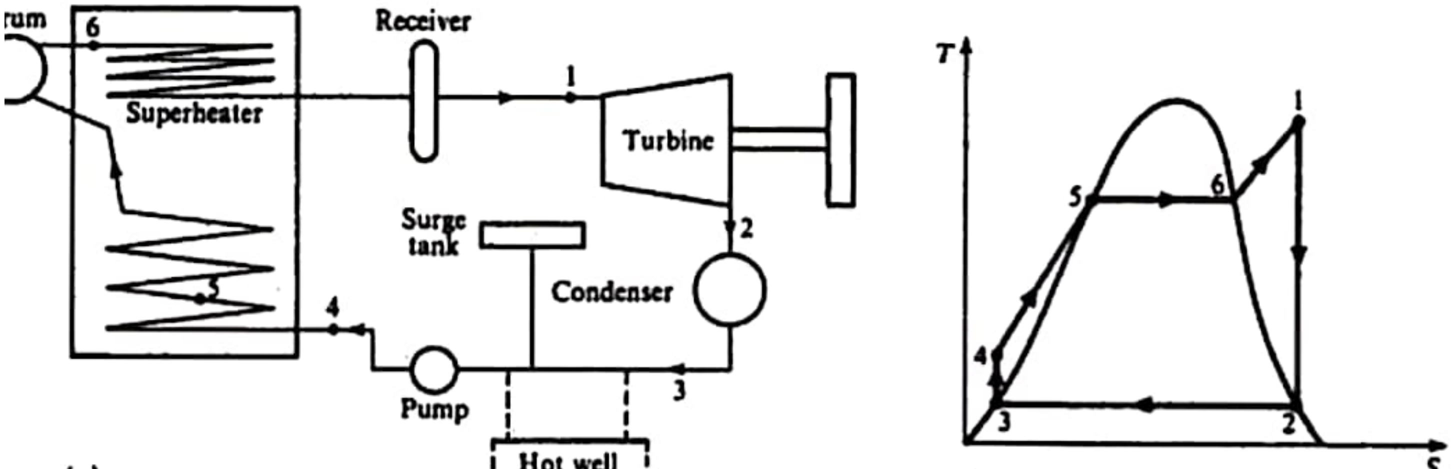
$$= \frac{793.6 - 4.2}{(2800 - 112) - 4.2} = 0.294$$

i.e. $$\text{Cycle efficiency} = 29.4\%$$

$$\text{Work ratio} = \frac{-W_{12} - \text{pump work}}{-W_{12}} = \frac{793.6 - 4.2}{793.6} = 0.995$$

Also

$$\text{ssc} = \frac{1}{793.6 - 4.2} = 0.001\,267 \text{ kg/kW s} = 4.56 \text{ kg/kW h}$$



(a) (b)

Example 8.2 Compare the Rankine cycle performance of Example 8.1 with that obtained when the steam is superheated to 500 °C. Neglect the feed-pump work.

Solution From tables, by interpolation, at 42 bar:

$$h_1 = 3442.6\ \text{kJ/kg} \quad \text{and} \quad s_1 = s_2 = 7.066\ \text{kJ/kg K}$$

Using equation (4.10)

$$s_2 = s_{f_2} + x_2 s_{fg_2} \quad \text{therefore } 0.391 + x_2 8.13 = 7.066$$

i.e. $\quad x_2 = 0.821$

Using equation (2.2)

$$h_2 = h_{f_2} + x_2 h_{fg_2} = 112 + (0.821 \times 2438) = 2113\ \text{kJ/kg}$$

From tables:

$$h_3 = 112\ \text{kJ/kg}$$

Then, using equation (8.2)

$$-W_{12} = h_1 - h_2 = 3442.6 - 2113 = 1329.6\ \text{kJ/kg}$$

Neglecting the feed-pump term, we have

$$\text{Heat supplied} = h_1 - h_3 = 3442.6 - 112 = 3330.6\ \text{kJ/kg}$$

Using equation (8.9)

$$\text{Cycle efficiency} = \frac{h_1 - h_2}{h_1 - h_3} = \frac{1329.6}{3330.6} = 0.399 \text{ or } 39.9\%$$

Also, using equation (8.14)

$$\text{ssc} = \frac{1}{h_1 - h_2} = \frac{1}{1329.6} = 0.000\,752\ \text{kg/kW s} = 2.71\ \text{kg/kW h}$$

Example 8.3 Calculate the new cycle efficiency and specific steam consumption if reheat is included in the plant of Example 8.2. The steam conditions at inlet to the turbine are 42 bar and 500 °C, and the condenser pressure is 0.035 bar as before. Assume that the steam is just dry saturated on leaving the first turbine, and is reheated to its initial temperature. Neglect the feed-pump term.

Solution The cycle is shown on a T–s diagram in Fig. 8.14.

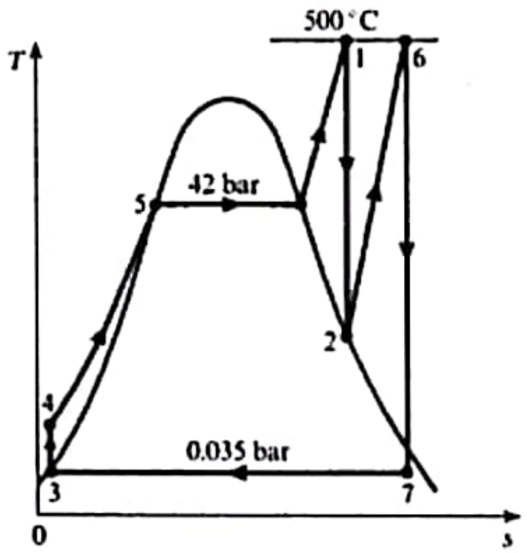


Fig. 8.14 T–s diagram for Example 8.3

It is convenient to read off the values of enthalpy from the h–s chart, i.e. $h_1 = 3442.6$ kJ/kg; $h_2 = 2713$ kJ/kg (at 2.3 bar); $h_6 = 3487$ kJ/kg (at 2.3 bar and 500 °C); $h_7 = 2535$ kJ/kg.

From tables

$$h_3 = 112 \text{ kJ/kg}$$

Then $\quad$ Turbine work $= (h_1 - h_2) + (h_6 - h_7)$

$$= (3443 - 2713) + (3487 - 2535)$$

i.e. $\quad$ Turbine work $= 1682$ kJ/kg

Heat supplied $= (h_1 - h_3) + (h_6 - h_2)$

$$= (3443 - 112) + (3487 - 2713)$$

i.e. $\quad$ Heat supplied $= 4105$ kJ/kg

therefore

$$\text{Cycle efficiency} = \frac{1682}{4105} = 0.41 \text{ or } 41\%$$

Also $\quad$ $\text{ssc} = \dfrac{1}{1682} = 0.000\,595$ kg/kJ

i.e. $\quad$ ssc $= 2.14$ kg/kW h